

2.- Creación de contenedores Simples

En esta actividad crearemos contenedores a partir de imágenes públicas de Docker Hub. Para verificar que los contenedores fueron creados correctamente y que sus puertos están bien configurados, utilizaremos **Table Plus** para establecer las conexiones. Usaremos las Siguietes Imágenes:

- MariaDB: jammy
- Postgres:16
- Redis:7

Creación de contenedor MariaDB: jammy

```
docker container run -d \  
> -p 3306:3306 \  
> --name MARIADB-V1 \  
> -e MARIADB_USER=jesuz \  
> -e MARIADB_PASSWORD=qwerty123 \  
> -e MARIADB_ROOT_PASSWORD=123 \  
> mariadb: jammy  
e746b3d173a3fa5c178bbae6704ea0d2f350adbac21008a3f83ea945f0b3da0f
```

Creación de contenedor Postgres:16

```
docker container run -d \  
> -p 5432:5432 \  
> --name POSTGRESDB-V1 \  
> -e POSTGRES_USER=jesuz \  
> -e POSTGRES_PASSWORD=qwerty123 \  
> postgres:16  
36b228589c676c1c69d5a4f5b64f56599711c370aec591d5e8ca42749afa7f70
```

Creación de contenedor Redis:7

A diferencia de los dos contenedores anteriores en este caso no contábamos con la imagen de redis descargada así que docker descarga la imagen y luego de ello crea el contenedor.

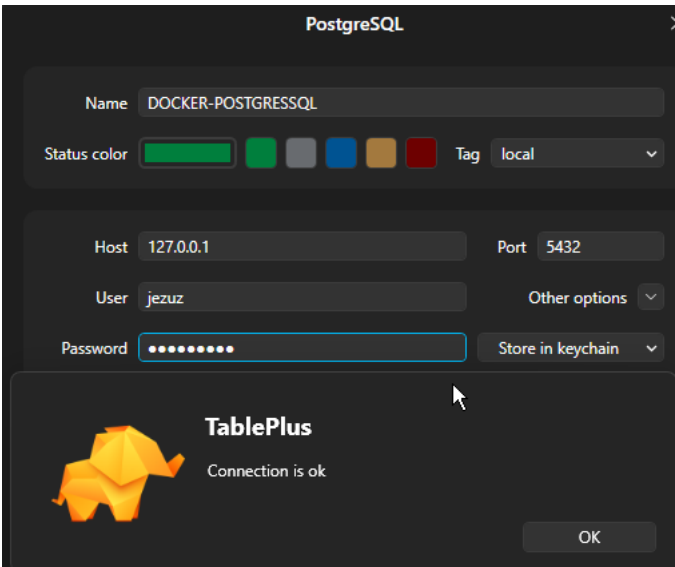
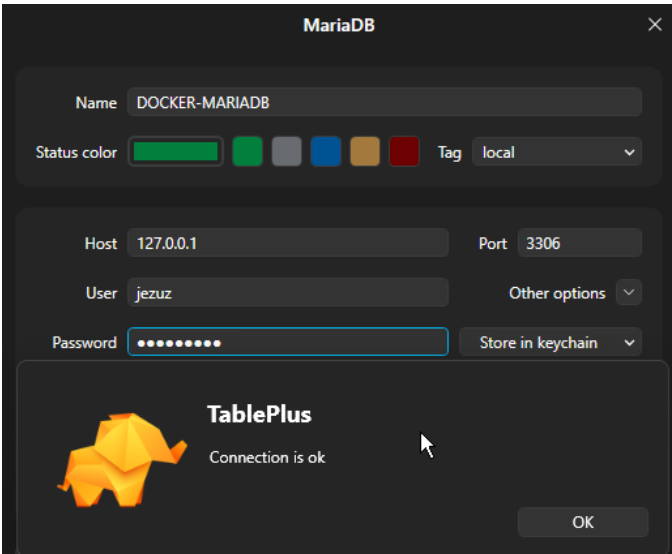
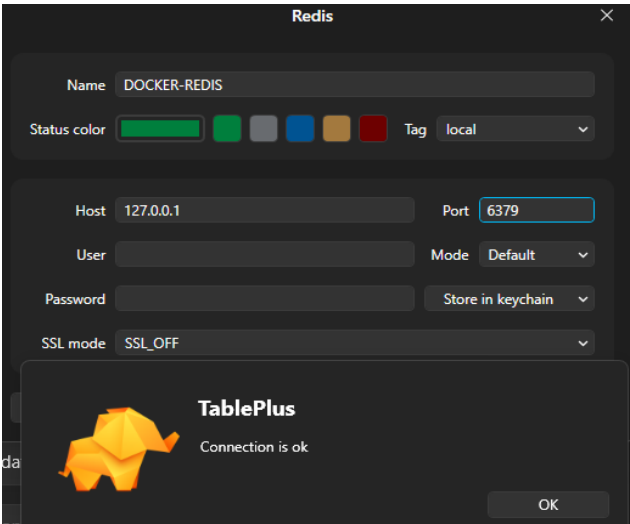
```
docker container run -d \  
> -p 6379:6379 \  
> --name REDISDB-V1 \  
> redis:7  
Unable to find image 'redis:7' locally  
7: Pulling from library/redis  
f4dd325b0baf: Pull complete  
a8ac7f6c67ab: Pull complete
```

f4f69793eb55: Pull complete
3f4e1fadfed1: Pull complete
4c75d4fe7471: Pull complete
4f4fb700ef54: Pull complete
d4044243cab2: Pull complete
e78e6c00c9a7: Download complete
2e752cc8487b: Download complete
Digest:
sha256:71da9275c5f3fcb97d0fa0c8c5b36cc995327265420f17a04bfd544f458059f7
Status: Downloaded newer image for redis:7
d9f399f27d32926c48f700aee1f323ec76d67096541ce3cb072b5e52fbf85c64

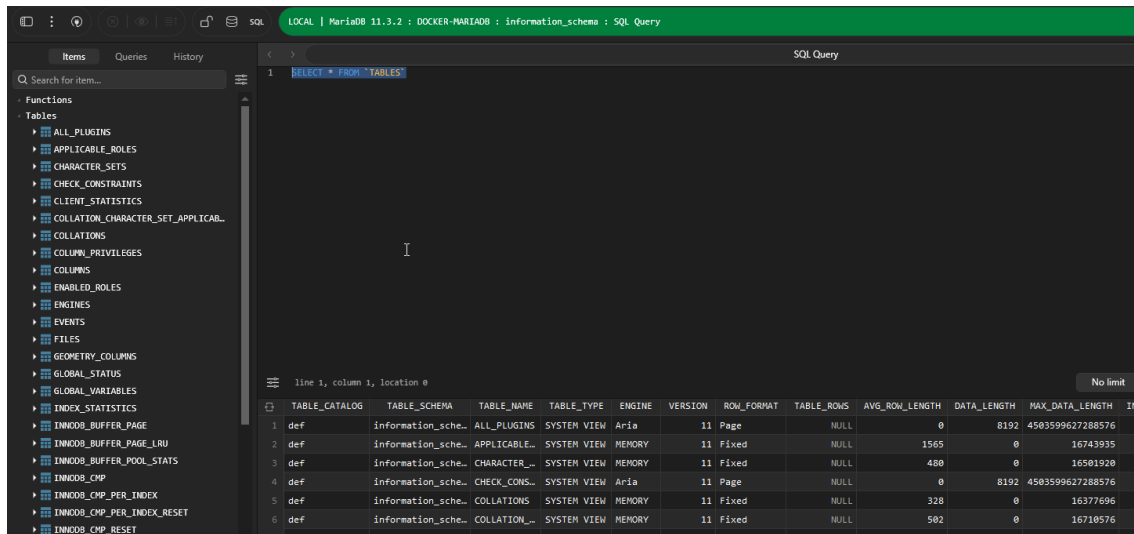
Verificamos si los contenedores fueron creados con éxito

☐	Name	Container ID	Image	Port(s)	CPU (%)	Memory usag...	Memory (%)	Disk read/write	Network I/O	PIDS	Last started	Actions
☐	MARIADB-V1	e746b3d173a3	mariadb:jan	3306:3306 ↗	0.02%	104.7MB / 15.6G	0.66%	71.8MB / 47.4ME	1.95KB / 126B	7	12 minutes ago	   
☐	POSTGRESDB-V1	36b228589c67	postgres:16	5432:5432 ↗	0%	39.32MB / 15.6G	0.25%	57.1MB / 57.9ME	998B / 126B	6	9 minutes ago	   
☐	REDISDB-V1	d9f399f27d32	redis:7	6379:6379 ↗	0.22%	3.28MB / 15.6GB	0.02%	111KB / 0B	872B / 126B	6	4 minutes ago	   

Verificamos si los contenedores tienen los puertos bien configurados



Realizamos la inserción de algunos Datos en el contenedor DOCKER-MARIADB



SQL Query

```
1 SELECT * FROM TABLES;
```

line 1, column 1, location 0

	TABLE_CATALOG	TABLE_SCHEMA	TABLE_NAME	TABLE_TYPE	ENGINE	VERSION	ROW_FORMAT	TABLE_ROWS	AVG_ROW_LENGTH	DATA_LENGTH	MAX_DATA_LENGTH	INDEX_SIZE
1	def	information_sche...	ALL_PLUGINS	SYSTEM VIEW	Ar1a	11	Page	NULL	0	8192	4503599627288576	16743935
2	def	information_sche...	APPLICABLE...	SYSTEM VIEW	MENORY	11	Fixed	NULL	1565	0	16501920	16718576
3	def	information_sche...	CHARACTER_...	SYSTEM VIEW	MENORY	11	Fixed	NULL	480	0	16501920	16718576
4	def	information_sche...	CHECK_CONS...	SYSTEM VIEW	Ar1a	11	Page	NULL	0	8192	4503599627288576	16718576
5	def	information_sche...	COLLATIONS	SYSTEM VIEW	MENORY	11	Fixed	NULL	328	0	16501920	16718576
6	def	information_sche...	COLLATION_...	SYSTEM VIEW	MENORY	11	Fixed	NULL	502	0	16718576	16718576

Detener un contenedor

`docker container stop POSTGRESDB-V1`
`POSTGRESDB-V1`

Iniciar un contenedor detenido:

`docker container start POSTGRESDB-V1`
`POSTGRESDB-V1`

Eliminación forzada

`docker rm -f REDISDB-V1`
`REDISDB-V1`

Eliminación estándar

`docker container rm POSTGRESDB-V1`
`POSTGRESDB-V1`

Revisar logs de un contenedor

`docker container logs MARIADB-V2`

```
jezuz@DESKTOP-NQ62FUL:/mnt/c/Users/Jesus/Desktop/Galaxy Training/Docker/Examen-Final/02.-Contenedores-Simples$ docker container logs MARIADB-V2
2026-09-07 03:07:01+00:00 [Note] [Entrypoint]: Entrypoint script for MariaDB Server 1:11.3.2+maria-ubu2204 started.
2026-09-07 03:07:02+00:00 [Warn] [Entrypoint]: /sys/fs/cgroup///memory.pressure not writable, functionality unavailable to MariaDB
2026-09-07 03:07:02+00:00 [Note] [Entrypoint]: Switching to dedicated user 'mysql'
2026-09-07 03:07:02+00:00 [Note] [Entrypoint]: Entrypoint script for MariaDB Server 1:11.3.2+maria-ubu2204 started.
2026-09-07 03:07:02+00:00 [Note] [Entrypoint]: Initializing database files
2026-09-07 3:07:02 0 [Warning] mariadb: io_uring queue init() failed with errno 1
2026-09-07 3:07:02 0 [Warning] InnoDB: liburing disabled: falling back to innodb_use_native_aio=OFF

PLEASE REMEMBER TO SET A PASSWORD FOR THE MariaDB root USER !
To do so, start the server, then issue the following command:

'/usr/bin/mariadb-secure-installation'
```

Renombrar un contenedor (MARIADB-V1 → MARIADB-V2):

MARIADB-V1 a MARIADB-V2

docker container rename MARIADB-V1 MARIADB-V2

CONTAINER ID	IMAGE	COMMAND	CREATED	STATUS	PORTS	NAMES
e746b3d173a3	mariadb:jammy	"docker-entrypoint.s..."	2 hours ago	Up 2 hours	0.0.0.0:3306->3306/tcp, [::]:3306->3306/tcp	MARIADB-V2

```
jezuz@DESKTOP-NQ62FUL:/mnt/c/Users/Jesus/Desktop/Galaxy Training/Docker/Examen-Final/02.-Contenedores-Simples$
```

Inspeccionar un Contenedor

docker container inspect MARIADB-V2

```
jezuz@DESKTOP-NQ62FUL:/mnt/c/Users/Jesus/Desktop/Galaxy Training/Docker/Examen-Final/02.-Contenedores-Simples$ docker container inspect MARIADB-V2
[
  {
    "Id": "e746b3d173a3fa5c178bbae6704ea0d2f350adbac21008a3f83ea945f0b3da0f",
    "Created": "2026-09-07T03:07:00.881506954Z",
    "Path": "docker-entrypoint.sh",
    "Args": [
      "mariadb"
    ],
    "State": {
      "Status": "running",
      "Running": true,
      "Paused": false,
      "Restarting": false,
      "OOMKilled": false,
      "Dead": false,
      "Pid": 754,
      "ExitCode": 0,
      "Error": "",
      "StartedAt": "2026-09-07T03:07:01.317400705Z",
      "FinishedAt": "0001-01-01T00:00:00Z"
    }
  },
]
```